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Breaking Barriers: Inspiring Careers in the Energy Sector

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Abstract

Inspiring the future workforce and providing the means to develop the skills of the next generation is vital to the success of the energy sector. Through supporting a wide range of STEM activities worldwide including STEM programmes, events, and available scholarships, as an industry, we can work together to ensure that young people are equipped with the relevant knowledge and skills, enabling them to pursue a career within the energy sector and help to shape a sustainable energy mix for the future. With the landscape of the energy industry transitioning from primarily fossil fuels to renewable energy sources, it is important for industry to engage with young people globally to encourage interest and educate them on STEM subjects whilst showcasing the diverse future career opportunities within the energy sector in line with their interests and strengths.

A research project was commissioned with the aim of gaining insight and a unique analysis of the key barriers perceived by young people in pursuing a career in the energy industry. A total of 4,000 people aged between 18–30 participated in the survey, from the USA, UAE, Brazil, and Malaysia, viewed as four major global energy hubs. Participants were split equally between each of the four countries and between genders. The survey consisted of questions summarising the importance of the energy transition, familiarity with employment opportunities, the impact of the energy transition on the future, and what they view as both challenges and barriers, and motivators when considering a future career within the energy sector.

Key takeaways from the survey include that 67% of participants were interested in pursuing a career within the energy industry, and 76% indicated that they believe the energy transition is either ‘important’ or ‘very important’. Significantly, within the UAE, although 58.5% of participants were in support of transitioning to renewable energy, 38.2% were in support of maintaining fossil fuels for energy—the highest among all other countries surveyed. When examining what participants perceived as the biggest challenges to working in the energy sector, the top three results were: special qualifications or certifications, lack of knowledge about the energy sector, and education (STEM qualifications).

The results show that increasing awareness, and the provision of appropriate skills is crucial to ensuring a safe and skilled workforce is available for the future of the energy industry. Awareness of job opportunities, whilst also creating clear pathways for young people to both pursue new skills and utilise existing ones, is pivotal to achieving this goal.

Introduction

With substantial global initiatives driving a transition from traditional fossil fuels to sustainable energy solutions, the importance of understanding the motivations, barriers, and awareness of job opportunities within the energy sector held by the future workforce was identified. Funding of international STEM education programmes and initiatives are key drivers in inspiring the next generation to pursue careers in the energy industry. By connection young people with interactive and engaging STEM activities, we are able to educate young people around the opportunities available to them in the energy industry in a way that is relatable and relevant to them. STEM activities and programmes aim to excite, stimulate and inform, while bringing the future of energy to life in a personal way so that more teenagers and graduates can imagine themselves playing a part in creating the energy mix the world will need to tackle the challenges of climate change and net zero.

Statement of Theory and Definitions

Donald Super's Lifespan/Life-Space theory (1) emphasises lifestyle factors affecting career prospects throughout the lifecycle. With the energy industry currently shifting from fossil fuels to a more sustainable energy future, it is deemed vital to target those who fall within Super's exploration age category and early into establishment, when carrying out this research to gain understanding of their awareness of the changes in the labour market and future career opportunities.

This research refers to the energy industry, however further breaking this down into two categories: fossil fuels, referring to traditional oil and gas and renewables, including solar, wind, carbon capture and hydrogen.

Description and Application of Equipment and Processes

The report surveyed 4,000 participants from global energy hubs UAE, USA, Brazil and Malaysia. 1,000 participants were cited from each country and divided into two age groups, 18-23 and 24-30. This included an equal split of males and females. The survey consisted of the following questions. MumtazAli et Al (2), refer to a sample size for survey research as being required to be over 200.

1. How important/unimportant do you believe the energy transition from fossil fuels (oil/gas) to renewables (solar, wind, etc) energy is?
2. Which of the following statements best supports your view on the energy transition from fossil fuels to renewables?
3. How familiar are you with the employment opportunities in the energy sector (including oil/gas and renewables)?
4. When you think of the energy sector, what kinds of roles/jobs do you think of?
5. How much impact do you think the transition from fossil fuels to renewable energy sources will have on your career in the next five years?
6. How likely are you to consider working in, or continue working in, the energy sector?
7. Which of these are the biggest challenges for you when considering working in the energy sector?
8. Which of these most encourages you to consider working in the energy sector?

The survey aimed to identify participants awareness of career prospects and the future of the energy industry regarding transitioning from traditional oil and gas roles to those required to support a more sustainable energy future. Alongside establishing awareness, the report aimed to highlight the largest perceived barriers to these career prospects and what typically motivates the next generation when pursuing career opportunities.

Presentation of Data and Results

Q1. How important/unimportant do you believe the energy transition from fossil fuels (oil/gas) to renewables (solar, wind, etc) energy is?

The options available for selection ranged from very important, important to not important and not important at all.

Table 1—Importance of the energy transition to young people.

	Average (rating between 1-10)	Top 2 positive answers	Bottom 2 answers
ALL	5	76%	3%
UAE	5.11	77%	2%
USA	4.74	64%	6%
Brazil	5.44	88%	3%
Malaysia	5.03	74%	2%

The results of the survey highlighted that regardless of global location, the next generation widely understands the importance of the energy transition from fossil fuels to renewables. Although greater recognition was found within Brazil than the other three countries surveyed. This could be due to historic emphasis placed on fossil fuels from the UAE, USA and Malaysia.

Q2. Which of the following statements best supports your view on the energy transition from fossil fuels to renewables?

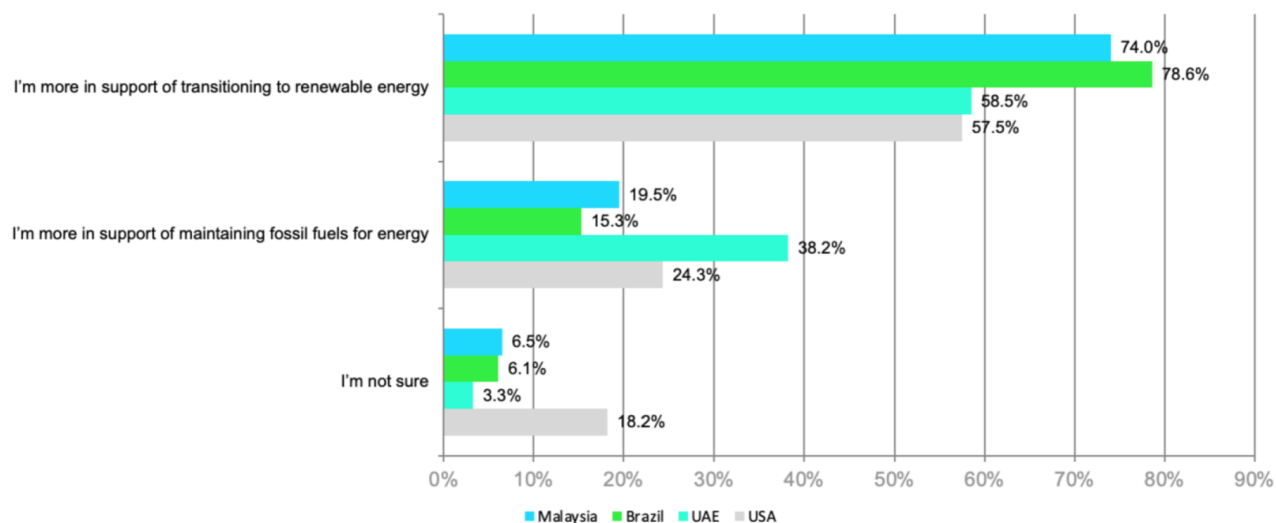


Figure 1—Support for fossil fuels vs renewables.

Following on from the trend set in question 1, Brazil was highlighted as the most supportive in transitioning toward renewable energy, with the UAE placing the most support on the importance of maintaining fossil fuels. Referring to Super's Lifespan/Life-Space theory this is likely due to the awareness created in early life that the country's economic security has been largely shaped by fossil fuels historically. This is also aligned with the countries emphasis on the requirement to balance the use of both fossil fuels and renewables, as opposed to entirely phasing out fossil fuels.

Q3. How familiar are you with the employment opportunities in the energy sector (including oil/gas and renewables)?

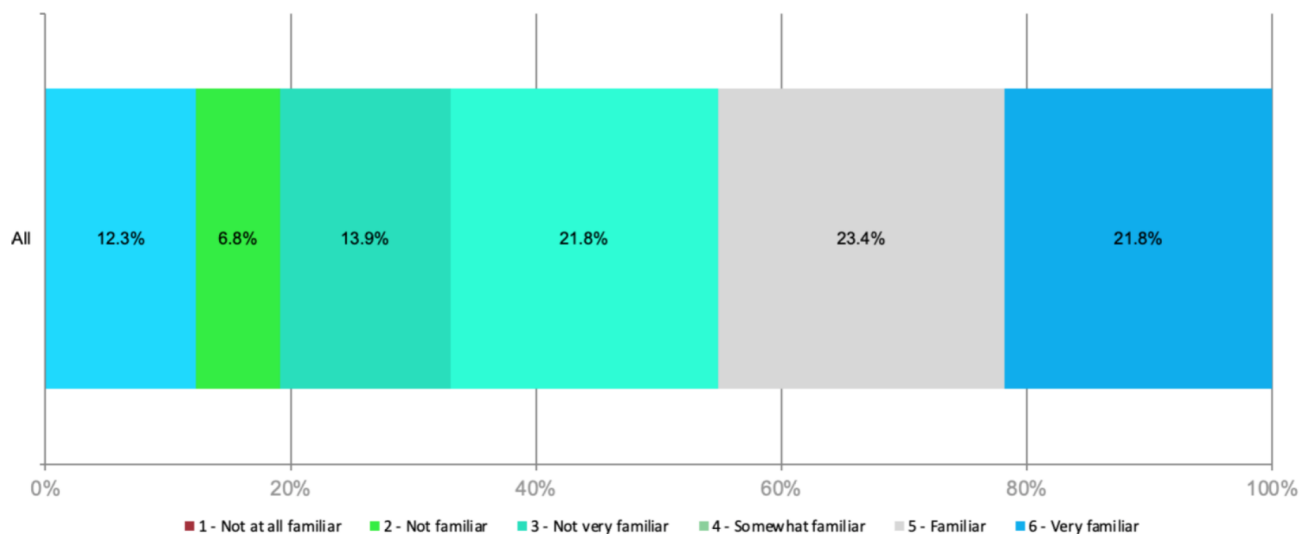


Figure 2—Familiarity with employment opportunities in the energy sector.

Regarding familiarity with employment opportunities in the energy sector, a combined 45% of respondents indicated a high level of familiarity, choosing ‘Familiar’ or ‘Very familiar’. This suggests a substantial proportion of the sample has a good understanding of job prospects available in this industry. On the opposite end of the spectrum, 19% of participants reported low familiarity of career pathways available to them in the industry, selecting ‘Not at all familiar’ or ‘Not familiar’. The majority of respondents fell in the middle categories, with 14% ‘Not very familiar’ and 22% ‘Somewhat familiar’, indicating a moderate level of awareness. The average rating across all participants was 4.03, which leans towards the ‘Somewhat familiar’ category, reflecting a general awareness of the energy sector's employment opportunities among the surveyed group.

Fig 3 has been further broken down into individual country responses, in fig 4 below.

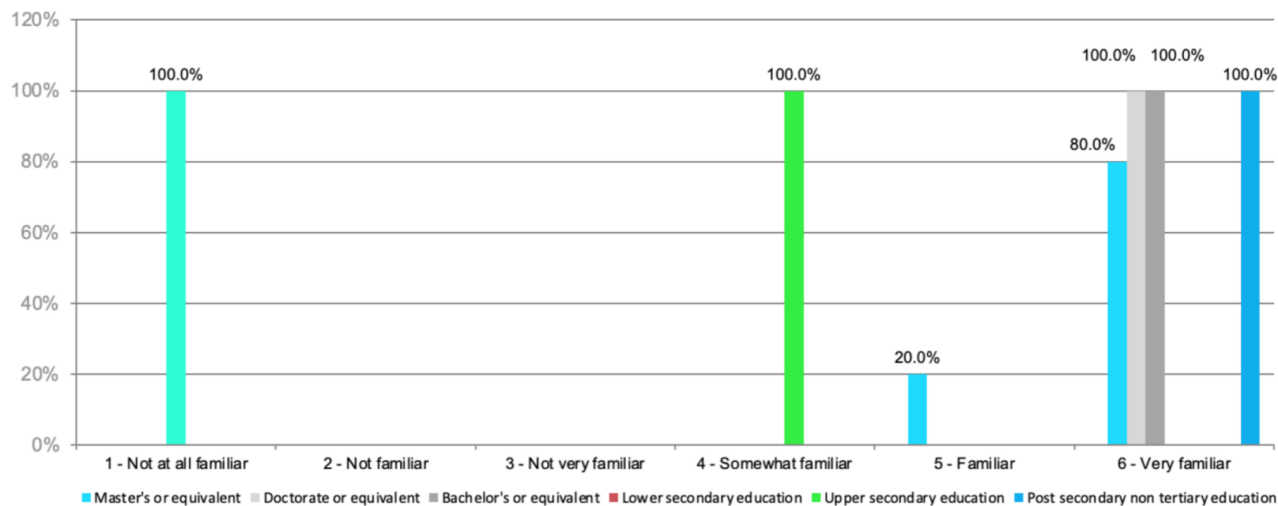


Figure 3—Familiarity with employment opportunities in the energy sector, by education status.

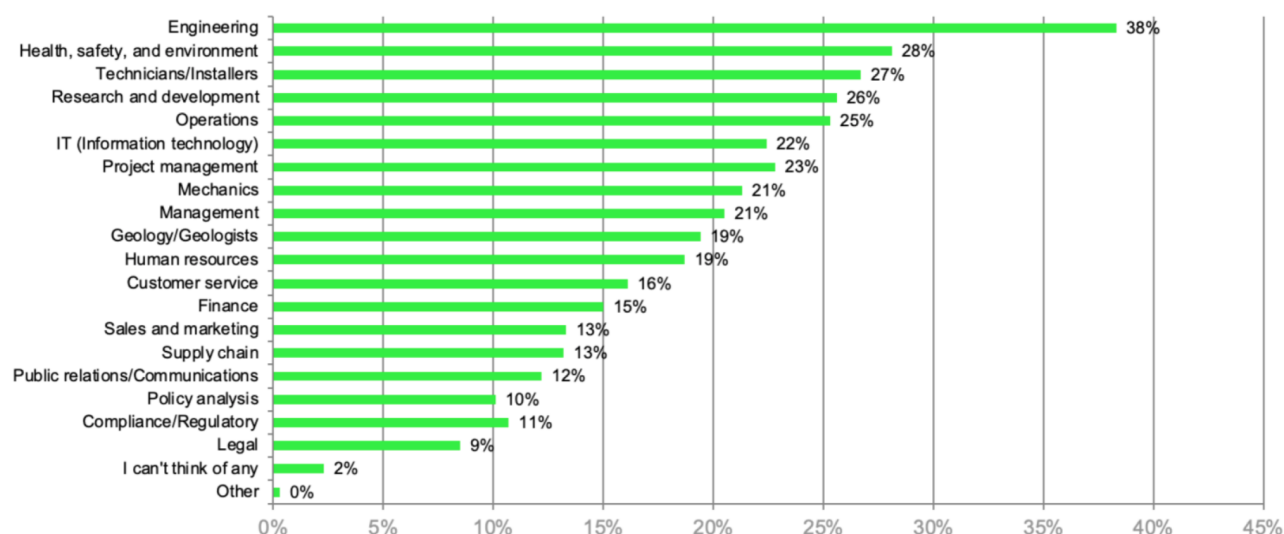


Figure 4—Awareness of roles within the industry.

Table 2—Familiarity with employment opportunities in the energy sector, by country.

	Average (rating between 1-10)	Top 2 positive answers	Bottom 2 answers
ALL	4	45%	19%
UAE	5	64%	8%
USA	4	40%	24%
Brazil	3	33%	31%
Malaysia	4	44%	13%

The survey results indicate varying levels of familiarity with employment opportunities in the energy sector across different countries. Participants from the UAE reported the highest levels of familiarity, with 64% of respondents indicating they are ‘Familiar’ or ‘Very familiar’. In contrast, Brazil showed the highest percentage of participants who are ‘Not at all familiar’ or ‘Not familiar’ with 31%. The USA presented a more balanced distribution across all levels of familiarity, with the highest percentage (22%) being ‘Very familiar’. Malaysia had the highest percentage of ‘Somewhat familiar’ responses at 28%. The average familiarity rating was highest in the UAE at 4.7, followed by Malaysia at 4.13, the USA at 3.83, and finally Brazil at 3.45. These results suggest that the UAE participants have a relatively better understanding of the energy sector's employment landscape, while there is a greater lack of awareness among participants in Brazil.

The survey results effectively emphasize the effect that education has on the understanding of career opportunities within the energy sector. This supports Super's theory regarding those within the explorative age category and obtaining an understanding of potential future careers. To ensure that the next generation are appropriately equipped for a future career in the energy industry it is therefore vital that the information is provided to them throughout their education journey.

Q4. When you think of the energy sector, what kinds of roles/jobs do you think of?

Participants were asked about what roles they associate most with the energy sector, with the most commonly cited job being ‘Engineering’, which was mentioned by 38% of respondents. This was then followed by ‘Health, safety, and environment’ (28%) and ‘Technicians/Installers’ (27%). Other prominent roles, each cited by over a quarter of respondents, were ‘Research and development’ (26%) and ‘Operations’ (25%). In contrast, roles such as ‘Legal’, ‘Policy analysis’, and ‘Compliance/Regulatory’ were

less commonly associated with the energy sector, each receiving 10% or less in responses. Notably, most participants could think of at least one role in the energy sector, with only 2% stating ‘I can't think of any’. The results indicate a strong association of the energy sector with technical and hands-on roles, as well as management and safety-related positions.

Q5. How much impact do you think the transition from fossil fuels to renewable energy sources will have on your career in the next five years?

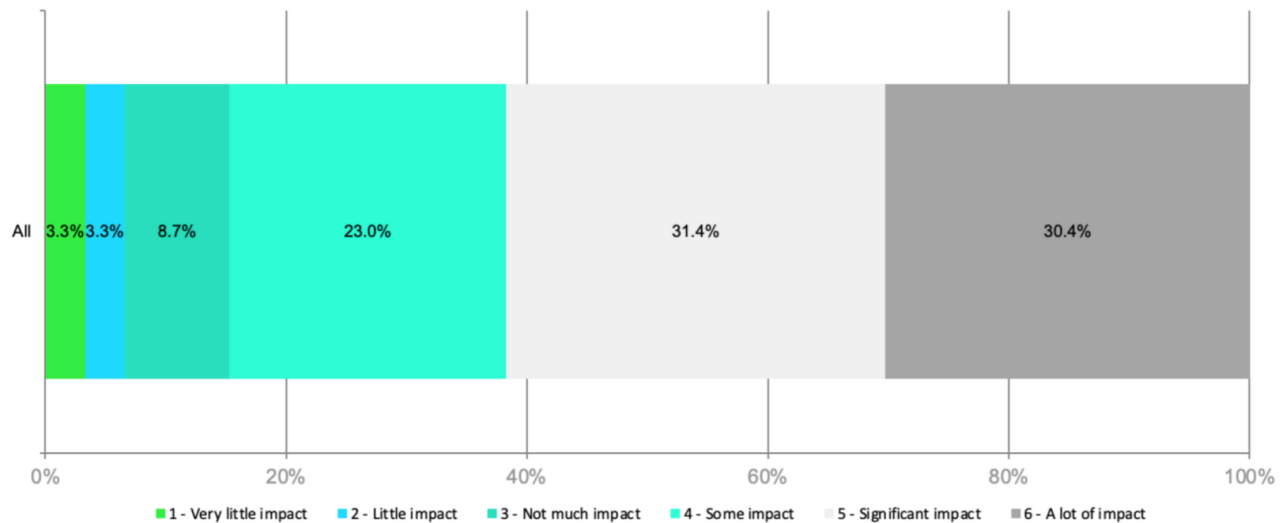


Figure 5—Awareness of energy transition impact on career prospects.

Regarding the impact of transitioning from fossil fuels to renewable energy on their careers in the next five years, a combined 62% of respondents anticipate a significant, or a lot of impact (31% and 30% respectively). A smaller segment, 23%, expects some impact, while the remaining 15% perceive less impact, with 9% indicating not much impact, and 6% collectively viewing little to very little impact (3% for each category). The average rating of 4.67 suggests a general expectation of a positive impact, with the majority of participants leaning towards the higher end of the impact scale emphasising an awareness of the energy transition and what it involves for the industry.

Q6. How likely are you to consider working in, or continue working in, the energy sector?

Table 3 below breaks down the statistics outlined in figure 6 and further details the responses received by country.

Table 3—Likelihood of a future career in the energy industry by country.

	Average (rating between 1 – 10)	Top 2 positive answers	Bottom 2 answers
ALL	4	52%	14%
UAE	5	67%	5%
USA	4	40%	22%
Brazil	4	43%	22%
Malaysia	5	59%	5%

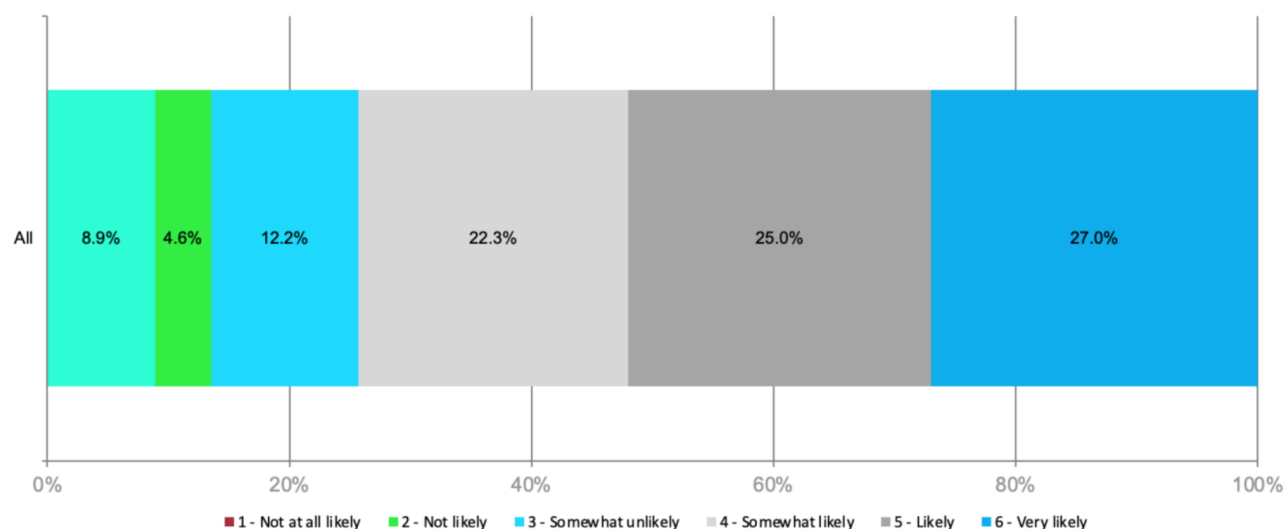


Figure 6—Likelihood of a future career in the energy industry.

The next table, [table 4](#) then displays what percentage, based on age category, concluded that they were either not likely or not likely at all to consider working, or continue to work in the energy sector.

Table 4—Likelihood of a future career in the energy industry by age.

	Average (rating between 1 – 10)	Bottom 2 answers
ALL	4	14%
18-24 years old	4	17%
25-30 years old	5	11%

The statistics outlined within [figures 6](#), and [tables 3](#) and [4](#) show that there is enthusiasm and interest from young people to explore a future in the energy industry whether that be by continuing to work in the industry or entering the industry. The remainder of the research outlines what the barriers and challenges, and key motivators are to the participants to explore this further.

Q7. Which of these are the biggest challenges for you when considering working in the energy sector?

[Figure 7](#) is further broken down in [figure 8](#) by age group.

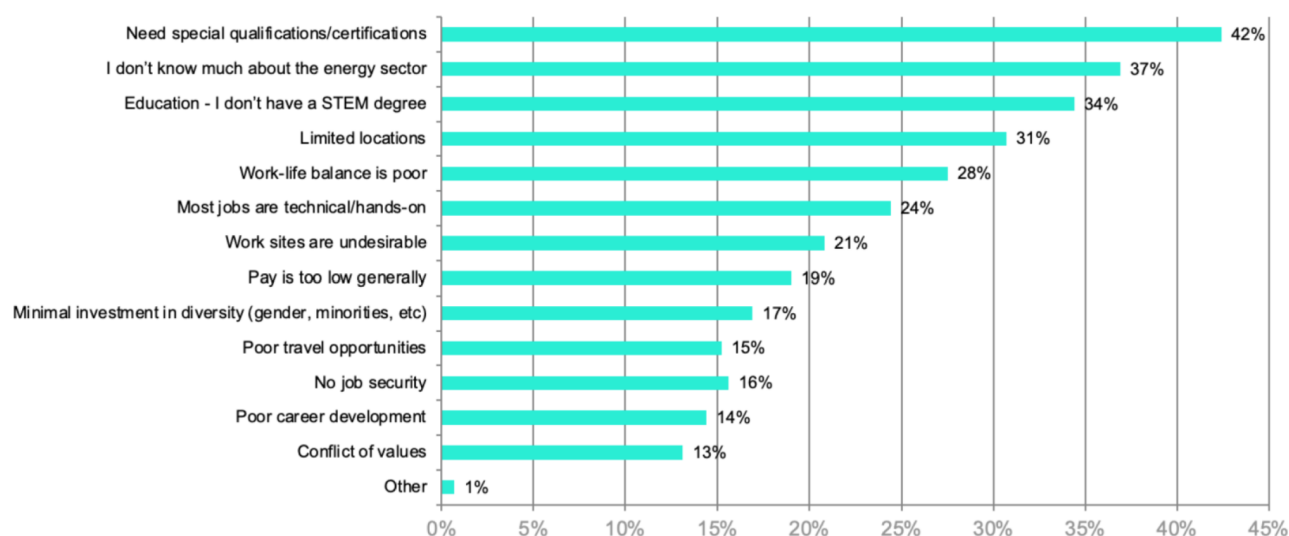


Figure 7—Biggest challenges and barriers to working in the energy sector.

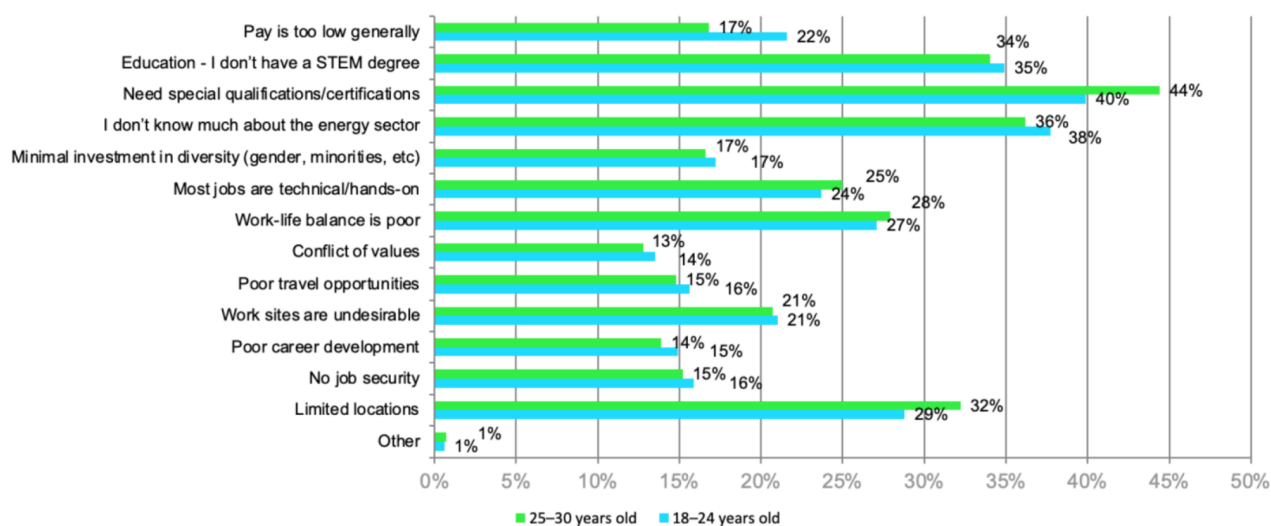


Figure 8—Biggest challenges to working in the energy sector by age group.

Lastly, these challenges and barriers have been divided by men and women, in figure 9.

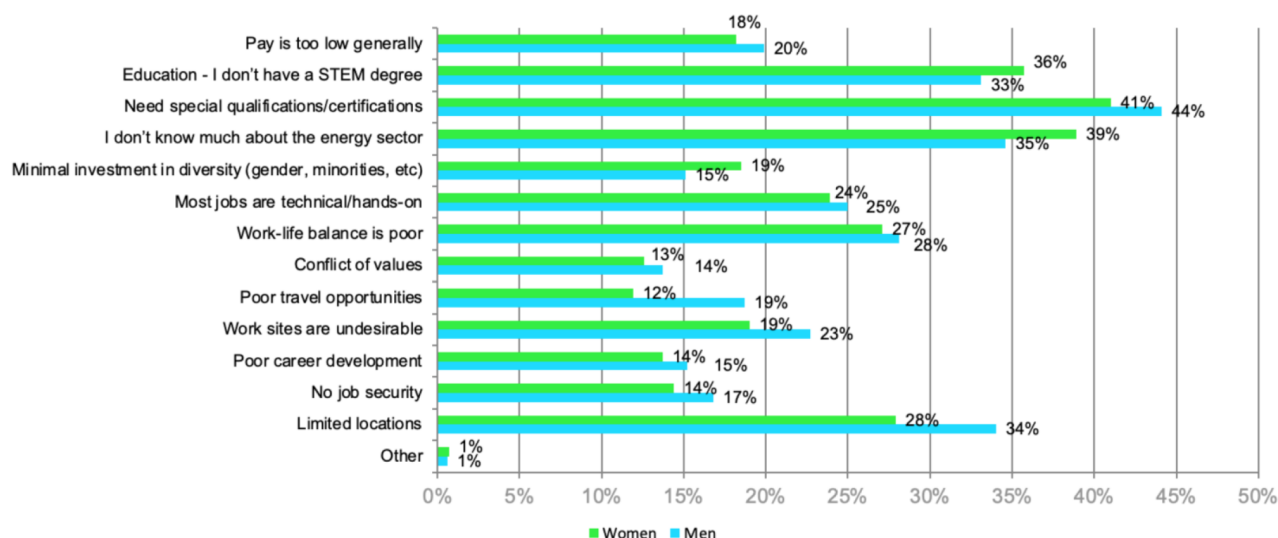


Figure 9—Biggest challenges and barriers to working in the energy sector by gender.

It is clear from figures 7, 8 and 9 that regardless of gender and age, special qualifications and certifications are viewed as the largest barrier in regard to pursuing a career in the energy sector. Nearly half of participants, particularly those aged 18-24 view this as the largest challenge. From this it is evident that there is substantial work that can be done to lower this statistic, by ensuring that suitable qualifications and certifications are available, encouraged and that the correct skills and any transferable skills and qualifications are identified and shared with the next generation of the workforce.

Following special qualifications and certifications, the next highest barrier identified is a lack of knowledge about the sector. Again, this is seen by both age categories and genders. Whilst the future workforce is currently in the explorative phase and/or early stages of their career, there is a gap identified here in the information which is currently available to ensure that they feel equipped to make the decision to both enter the energy sector, but additionally in a capacity that would interest them. Although earlier in the survey we saw general awareness around the industry, the responses here identify that they personally do not feel as though they have all of the knowledge required for a future in the industry.

Lastly, and in keeping with the tone of the first two barriers, the third highest noted is that of an education in STEM related subjects. This is further aligned with the findings in figure 6, where the most thought of role relevant to the energy industry is that of engineering. Again, although the participants seem to have a general awareness, they are lacking knowledge about the breadth and variety of opportunities available to them.

Q8. Which of these most encourages you to consider working in the energy sector?

Lastly, the participants were asked what would motivate them to consider continuing or pursuing a career in the energy sector. Responses are noted in figure 10.

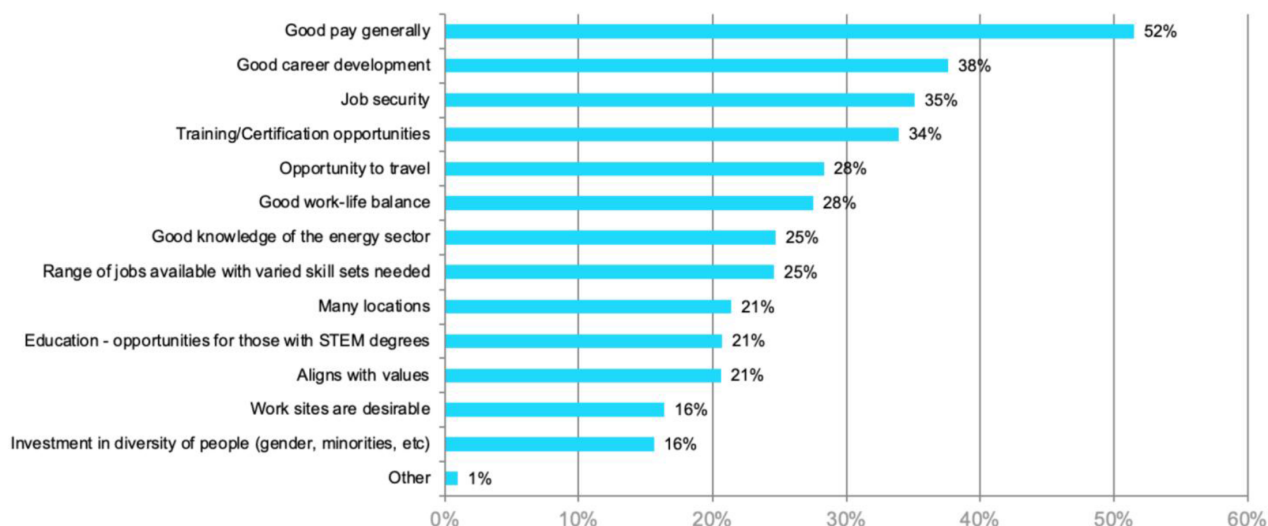


Figure 10—Motivators to pursuing a career in energy.

Given that 67% of participants earlier showed interest in pursuing a career in the energy industry, it is important to understand what would drive them too so. Results show that over half of the participants main motivation is financial based. This highlighted that if the barriers identified, around qualifications, certifications, knowledge/understanding and education can be mitigated, they would be willing to move in to the energy industry, as long as the financial situation was suitable.

The second and third most popular motivators identified by participants is career development and job security, which shows a willingness to grow in the industry and maintain a lifelong career path. Closely, behind job security was site training and certification opportunities, which further supports that participants would be interested in progressing within the industry if the appropriate qualifications and certifications were clear and available to them.

Conclusions

The report highlights the importance of reducing entry barriers to the industry by improving educational resources, job awareness, clear pathways, and career stability. These findings make it clear that industry must provide the correct tools to allow young people to fulfil their ambition in energy careers. By mitigating these barriers, industry can contribute towards ensuring the success of the sector by upskilling the future and current workforce as the energy transition picks up pace.

Increasing awareness of job opportunities, across all potential roles within the industry is crucial whilst the future workforce is in the explorative stage of their career life. The statistics show that the interest and willingness to pursue a long-term future in the industry is there. However, access the specific details and required skills, knowledge and qualifications is the largest gap.

Investment into STEM initiatives aims to close these gaps and provide both industry and the next generation with the information required for us to move towards a future safe and skilled energy workforce. Regional STEM activities are an effective way to engage with young people and pique their interest in the roles and opportunities available through and doing so in a relatable way that is insightful and engaging. Through STEM initiatives, industry can provide dynamic and relevant content aimed at raising awareness and driving engagement amongst the next generation, showcasing the diverse career opportunities available within the sector, and identifying current and future jobs across a wide range of energy-related fields, both within and beyond traditional STEM focuses.

Alongside side this, governments and industry leaders are engaging with industry to ensure that the skills required to support this movement are both clear and available to those who wish to seek them. The

adaptation of available Products and Qualifications to support the movement and combat these barriers, which now includes energy transition qualifications, Standards and open learning courses is also vital to this mission to mitigate the barriers for the next generation to enter the energy sector.

Acknowledgements

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